

Data for Large Language Models

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

总论：模型只能学习训练数据里的知识、推理模式、语言分布和偏差，所以 data is the key → 数据阶段：pre-training 造 base model，mid-training 强化能力，post-training 把模型调成会听指令的 instruct model → 数据来源：Wikipedia 高质量但有代表性偏差；web/Common Crawl 巨大但脏；GitHub 对代码与结构化推理重要但受 license 约束 → 选择目标：data selection 同时追求 performance、data efficiency、selection efficiency、evaluation integrity 和降低 toxicity/bias → 过滤流水线：language filtering、heuristics、quality filtering、domain filtering、deduplication、toxicity/PII filtering、data mixing 共同把 raw data 变成 corpus → 去重机制：exact matching 处理完全重复，MinHash 用紧凑签名近似 Jaccard similarity 来抓 near-duplicates → FineWeb 案例：从 Common Crawl 出发系统调试 filters 和 dedup，证明 quality trumps quantity → FineWeb-Edu 案例：用强 LLM 生成教育价值 synthetic labels，再训练轻量 classifier 低成本筛 15T tokens → Post-training：instruction data、synthetic instruction data 和 alignment data 把续写器变成任务执行者 → 伦理法律：representation、toxicity、PII、copyright、license 和 terms of use 决定数据工程不是纯技术问题

Data is the key

Data determines capabilities

- Model learns only what's in the training data (knowledge, reasoning patterns, biases)

Data quality requires human effort

- Most resource intensive and valuable component of modern AI
- Requires sustained investment into human expertise and careful curation
- Long tail of problems: domains, edge cases, quality issues

Competitive Differentiator

- Proprietary datasets (GPT-4, Claude) provide performance edge
- Architectural innovations shared in papers, system engineering can be reverse-engineered – datasets give sustained advantage
- Reasons for secrecy: (i) competitive dynamics and (ii) copyright liability

这节课一句话

L05 的核心不是“数据越多越好”，而是：LLM 的能力、偏差、成本、评测可信度和法律风险，都被一条从数据来源到筛选、去重、后训练、合成数据与伦理治理的 data pipeline 共同塑形。

1. Data is the key: 模型能力是数据分布的影子

Data is the key

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Competitive Differentiator

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人话直觉

把 LLM 想成一个极其勤奋、但从不亲眼见世界的学生。它所有知识、语气、推理套路、偏见和盲区，都来自课本和练习册；如果练习册里没有某个领域、某种方言、某类边界案例，它就很难凭空长出来。数据不是训练前的后勤材料，而是模型能力的地基。

精确定义 / 机制： Slide 3 的三句话构成整节课的总论：data determines capabilities；data quality requires human effort；data is a competitive differentiator。预训练目标表面上只是预测下一个 token，但模型学到的统计模式包含事实知识、语言习惯、推理模板、价值偏向和错误关联。因此数据质量不是单一指标：它同时涉及覆盖面、准确性、多样性、领域平衡、边界案例、噪声、重复、版权和安全。为什么大公司常不公开训练数据？因为架构会被论文扩散，系统工程可被逆向，持续维护的高质量专有数据却很难复制；同时公开来源也可能暴露 copyright liability。

考试问法：如果考“why data is key”，不要只答“数据多所以好”。高分答案按四层写：模型只能从数据中学习；数据质量和覆盖面决定能力与偏差；清洗和策展需要人类专家与长期投入；专有数据带来竞争优势但也带来版权风险。

2. 三种训练数据：base model 和 instruct model 不是同一种动物

Types of Data

Pre-training data – on raw text

Annealing or mid-training – more focus on particular capabilities, higher quality data

Post-training – focus on evaluation tasks, desired behaviour

consist of: instruction training, RL, alignment, task fine-tuning

Terminology:

Base model: after pre-training + annealing

Instruct model: after post-training

Each task has own data: translation, long context, reasoning etc. which we will look at in future lectures

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人话直觉

预训练像让学生读完整个图书馆，mid-training 像考前把某些专题强化刷题，post-training 像教他说人话、听任务、遵守课堂纪律。读书多不等于会按用户要求回答；会续写也不等于会帮忙。

精确定义 / 机制： Slide 4 把 LLM 数据分成三层。Pre-training data 通常是 raw text/code，目标是学习通用语言、世界知识和统计模式，产物是 base model。Annealing 或 mid-training 出现在预训练后期，用更高质量或更聚焦的数据强化特定能力，例如推理、长上下文、代码或领域知识。Post-training data 则面向 desired behaviour，包括 instruction training、RL、alignment 和 task fine-tuning，产物是 instruct model。这个区分很关键：base model 可能有知识但不一定知道“用户要我完成什么任务”；instruct model 通过 instruction-response、偏好或安全数据学习如何回答、拒答、解释和遵循格式。

考试问法： 如果考 pre-training vs post-training，先定义训练目标，再定义数据形式，最后说产物：pre-training 学通用分布得到 base model；post-training 学任务、偏好、安全和期望行为得到 instruct model。Mid-training 可作为中间桥梁：用高质量/能力导向数据继续塑形。

3. 数据来源：Wikipedia 是净水，Web 是海水，GitHub 是工具箱

Web Data

- The biggest and most freely available data source is the web
- How big is it? **HUGE**
 - Google search indexes about **400 billion web pages**
- Actual web far larger in the deep web – can be just institutional data that is behind a login eg. Amazon, but also dark web with deliberately encrypted data
- Spans a broad range of **domains, genres, languages** – world knowledge and linguistic knowledge that LLMs need
- Broad but not universal – overrepresents younger users and developed countries especially USA

<https://zyppy.com/seo/google-index-size/>

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人话直觉

训练 LLM 像给一座城市供水：Wikipedia 像过滤过的瓶装水，干净但量有限且产地偏；web data 像海洋，取之不尽但又咸又脏；GitHub 像一间巨大的工具房，里面有代码、工程习惯和问题分解方式，但每件工具都有许可证标签。

精确定义 / 机制：预训练语料通常混合多种来源。Wikipedia 质量高、结构清晰、多语言、Creative Commons、元数据丰富，几乎所有 LLM 都会用，但它存在地理和人口统计偏差，例如女性编辑比例低，某些地区和群体代表性不足。Web data 是最大、最自由可获得的数据源之一，覆盖世界知识、语言风格、领域和体裁；但它也过度代表年轻用户、发达国家、美国和英语互联网，并充满 boilerplate、广告、模板、重复、垃圾和错误。Common Crawl 的意义在于把原本只有大公司能大规模爬取的网页数据开放给研究者，但它只是 raw material，不是高质量训练集。GitHub 数据对编程能力和结构化推理重要，stars、issues、pull requests、项目活跃度和 license 都是筛选信号；The Stack 这类数据集强调 permissively licensed code，说明代码数据尤其不能忽略授权。

考试问法：如果题目问 pre-training data sources，可以按 Wikipedia / web/Common Crawl / GitHub 三类比较：优势、局限、为什么需要过滤。一定要写出“web broad but not universal”：规模大不代表公平覆盖，也不代表质量高。

4. Data selection: 把 raw data 变成训练语料的一串闸门

Pipeline



Albalak et al. (2024) A Survey on Data Selection for Language Models, TML

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人话直觉

不要把 data selection 想成一次性删除垃圾邮件，而要想成机场安检流水线：先看护照语言，再过尺寸和格式检查，再查危险物品，再防止同一个人重复排队，最后决定不同航班放多少人。每个闸门都可能救模型，也可能误伤好数据。

精确定义 / 机制：Slides 13–15 给出 data selection 的目标与 pipeline。总体目标是构造尽量接近测试/部署分布的 optimal dataset，同时减少成本、保护 evaluation integrity、减少 undesirable behaviours。常见目标包括 model performance、data efficiency、selection efficiency、evaluation integrity、reduce bias/toxicity。预训练过滤 pipeline 可拆成 language filtering、heuristics、data quality、domain specific、deduplication、toxic & explicit content、data mixing。Language filtering 用 URL 线索或 fastText 等分类器控制语言比例，但短文本、低资源语言、相似语言、方言和 code-switching 都难处理。Heuristic filtering 用字符数、词数、重复率、标点、符号比例等快速规则去除 boilerplate、错误信息和明显低质量文本，但它只能看表面统计，不能真正理解内容价值。Classifier-based quality filtering 则用参考高质量语料训练模型，判断样本是否接近目标分布，成本更高但更灵活。

考试问法：如果考“data selection pipeline”，按顺序列环节并各给一句作用。高分点是指出 trade-off：过滤能降成本、提质量、保护评测，但阈值和分类器也可能系统性排除方言、低资源语言或边缘群体内容。

5. Deduplication 与 MinHash: 别让模型把同一张传单背一万遍

MinHash

1. Convert Documents to Sets

Break each document into tokens (words, character n-grams, etc.)

Example: "the cat sat" → {the cat, cat sat}

2. Create Hash Signatures

Apply multiple hash functions to each element in the set

For each hash function, keep only the *minimum* hash value (hence "MinHash")

This creates a compact "signature" (e.g., 128 numbers) representing the document

3. Estimate Similarity

Compare signatures instead of full documents

The fraction of matching values in the signatures approximates the **Jaccard similarity** (overlap between sets)

If 64 out of 128 signature values match, similarity = 50%

- Documents with similar content will likely produce similar minimum hash values.
- The more hash functions you use, the more accurate the similarity estimate.
- Works on billions of documents (used by Common Crawl, Google, etc.)

Optional details: <https://medium.com/@omkarsoak>

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人话直觉

互联网里同一段文字会像传单一样到处被复制：新闻转载、模板页、导航栏、镜像站、旧 crawl 反复出现。如果模型反复看到同一张传单，它不是更博学，而是更容易浪费算力、记住噪声、在评测里靠背答案作弊。MinHash 就像给每篇文章做一枚压缩指纹，用指纹相似度判断它们是不是近亲。

精确定义 / 机制： Deduplication 的目的有三类：减少 dataset size 和训练资源，提高泛化，降低对低质量重复文本的过拟合和记忆风险。Exact matching 可按完全相同内容或 URL 删除重复；approximate matching 用于模板相同但细节略变的 near duplicates。直接比较所有文档代价不可接受，因为文档对数量随 n 近似平方增长，百万文档约有五千亿对。MinHash 的机制是：先把文档切成 tokens 或 character n-grams 的集合；对集合元素应用多个 hash functions；每个 hash function 只保留最小 hash value，形成固定长度 signature；比较两个 signatures 中匹配位置的比例，用来估计原集合的 Jaccard similarity。签名越长，估计越稳定；签名相似度高的文档很可能内容重叠大，可进入近重复删除。

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad \Pr[h_{\min}(A) = h_{\min}(B)] = J(A, B)$$

考试问法：如果考 MinHash，按三步答：sets → hash signatures → compare signatures。不要只说“用 hash 去重”；要明确它估计的是 Jaccard similarity，并解释为什么它比全文两两比较更适合 billions of documents。

6. FineWeb: 质量胜过数量，不是口号而是实验策略

Put in Practice: FineWeb

- Quality trumps quantity!
- FineWeb, a 15T token dataset of text sourced from 96 Common Crawl snapshots
- Principled strategy for choosing and tuning filtering heuristics that helped produce a small set of effective filters out of over fifty candidate filters
- How different deduplication strategies and granularities can impact performance
- Data quality using educational classifier

[Penedo et al. \(2024\) The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale, NeurIPS](#)

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人话直觉

FineWeb 像一个认真调味的厨师：Common Crawl 给的是一仓库食材，不是成品菜。它不满足于“多放点”，而是反复试哪几个过滤器真正让模型表现变好，哪种去重粒度会保留营养、去掉霉味。

精确定义 / 机制： FineWeb 是 L05 的核心案例之一：一个约 15T tokens、来自 96 个 Common Crawl snapshots 的 web text dataset。Slide 21 的主旨是 quality trumps quantity。它系统比较五十多个候选 filtering heuristics，最后选择少量有效过滤器；还研究不同 deduplication strategies 和 granularities 如何影响 performance。Slide 22 强调评估过滤策略很贵：可以人工检查 filtered/unfiltered data，也可以训练模型并在 benchmark 上测，但后者更可靠也更慢更昂贵。Slide 23 的 dedup 例子提醒我们：global dedup、individual snapshot dedup、跨 crawl 重复簇的处理都会改变 token 数和模型表现；去重不是越狠越好，而是要看是否去掉了无意义大重复，还是误删了有价值分布。FineWeb 因此体现的是现代数据工程方法论：提出候选过滤器，用实验验证，把质量、规模、成本和 downstream performance 绑在一起优化。

考试问法：如果考 FineWeb，标准答案是：从 Common Crawl 出发，15T tokens、96 snapshots；系统选择 filtering heuristics；比较 dedup strategies；用 educational classifier 做质量过滤；核心结论是 quality trumps quantity。可补充：过滤效果最好通过训练模型和 benchmark 评估，但代价高。

7. FineWeb-Edu: 用 synthetic labels 让强模型当标注老师

FineWeb-Edu

Use synthetic labels of **Educational Content** for training filters – getting at data quality but efficient

Below is an extract from a web page. Evaluate whether the page has a high educational value ... Points are accumulated based on the satisfaction of each criterion:

- Add 1 point if the extract provides some basic information relevant to educational topics, ...
- Add another point if the extract addresses certain elements pertinent to education ...
- Award a third point if the extract is appropriate for educational use and introduces key concepts relevant to school curricula...
- Grant a fourth point if the extract highly relevant and beneficial for educational purposes for a level not higher than grade school, exhibiting a clear and consistent writing style...
- Bestow a fifth point if the extract is outstanding in its educational value, perfectly suited for teaching either at primary school or grade school...

The extract: <EXAMPLE>
After examining the extract:
- Briefly justify your total score, up to 100 words.
- Conclude with the score using the format: "Educational score: <total points>"

Used Llama-3-70B-Instruct to score 460,000 sample documents from FineWeb

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人话直觉

如果让最贵的老师批改 15T tokens 作业，学校会破产；但可以先让老师批改 46 万份样本，教会一个便宜助教什么叫“有教育价值”，再让助教筛完整个仓库。FineWeb-Edu 的聪明之处就在这里。

精确定义 / 机制： FineWeb-Edu 展示了 synthetic data 不只可以生成 instruction，也可以生成 labels 来训练 data filters。流程是：先用 Llama-3-70B-Instruct 对 460,000 个 FineWeb sample documents 打 educational value 分数，评分 prompt 从 1 到 5 描述基本信息、教育相关性、课程适配、清晰度和教学价值；再在 Snowflake-arctic-embed-m embeddings 上训练 lightweight linear regression classifier；最后用这个分类器高效过滤 15T tokens，而不是直接用 LLM 扫全量数据。Slide 25 给出关键数字：直接用 LLM 成本会超过 \$1M；纳入阈值是 score ≥ 3 ；阈值在 reasoning benchmarks (MMLU、ARC) 和 general benchmarks (HellaSwag) 之间折中；最终从 15T tokens 中保留约 1.3T tokens 高质量 educational content。这个案例把“质量”从抽象评价变成可扩展的分类器决策。

$$\hat{s}(x) = w^T e(x) + b, \quad x \in \text{FineWeb-Edu} \iff \hat{s}(x) \geq 3$$

考试问法： 如果考 FineWeb-Edu 或 synthetic labels，答题顺序是：强 LLM 小规模打分 $\rightarrow$ embeddings + linear regression classifier $\rightarrow$ score threshold $\geq 3 \rightarrow$ 大规模过滤 15T tokens $\rightarrow$ 保留 1.3T educational tokens。高分点：说明它是 data quality filtering，也是 synthetic data 的一种用法。

8. Post-training 与 synthetic instruction data: 让模型从会续写变成会办事

Instructions

Limitations of pre-trained models:

No explicit notion of "what task am I being asked to do?"

Does not learn how to follow goals, answer questions or behave helpfully

Instruction: Summarize the following text in one sentence.

Response: The text explains how photosynthesis converts sunlight into energy for plants.

Instruction: Translate the sentence into French: "Good morning."

Response: Bonjour.

Instruction: List two benefits of regular exercise.

Response: Regular exercise improves cardiovascular health and reduces stress.

What instruction tuning adds

A supervised signal for **following commands** – what kind of behaviour is expected

Exposure to **many task types** via shared instruction format turns word predictor into system can generalize

Alignment with how humans naturally interact with models, and how to be helpful, and truthful

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人话直觉

预训练模型像一个读过全网的接龙高手：你给它半句话，它能往下编；但你让它“总结、翻译、列两点、拒绝危险请求”，它需要见过这种任务格式和期望答案。Instruction tuning 就是在教它：人类说这句话时，真正要你做什么。

精确定义 / 机制： Slide 30 指出 pre-trained models 的限制：没有明确 notion of what task am I being asked to do，也没有学会如何 follow goals、answer questions 或 behave helpfully。Instruction tuning 用 instruction-response pairs 提供监督信号，例如总结、翻译、列举好处，使模型接触许多 task types，并通过共享指令格式形成 zero-shot 或 cross-task generalization。Super-Natural Instructions 覆盖 1600+ NLP tasks，说明任务说明本身就是可训练对象。Synthetic instruction data 进一步扩展了 post-training 数据来源：self-generation + verification 适合数学、代码等可验证任务；distillation from stronger models 让强模型生成数据训练弱模型；evolutionary methods 如 Evol-Instruct/WizardLM 通过改写复杂化指令；procedural generation 适合结构化任务。Alpaca 用较低成本生成 52K instruction-response pairs，展示开源模型可以借助 stronger model data 获得较强指令跟随。

考试问法：如果考 instruction tuning 加了什么，要答：它加的是遵循任务和期望行为的 supervised signal，而不是新架构。若考 synthetic instruction data，要同时写用途和风险：低成本扩展、多样任务覆盖；但需要质量控制、ground truth 验证、混入 human data 防止多样性坍缩和 teacher bias 循环放大。

9. 数据伦理与法律：过滤器也是价值判断器

Representation

- Training data shapes whose voices, cultures, and perspectives models learn to reflect
- Underrepresentation can lead to bias, stereotyping, and exclusion of marginalized groups
- Overrepresentation of dominant groups may reinforce existing power imbalances
- Cultural and linguistic gaps can reduce model fairness, accuracy, and usefulness

Example: African American dialect
Prompt: Is this correct English? "She be finished her homework."
Response: No

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人话直觉

数据管线表面上是在删噪声，实际上也在决定谁的声音算“正常”、谁的语言算“错误”、谁的个人信息会被记住、谁的作品被拿去训练。一个坏过滤器不是简单少学一点，而可能把社会不平等压缩进模型参数。

精确定义 / 机制： L05 最后一段把 data engineering 推向 ethics and law。Toxic/explicit filtering 能减少模型生成有害内容，但也有副作用：模型可能不擅长识别这类文本，词表或分类器还可能误伤某些群体语言。PII 包括姓名、地址、电话、信用卡号、邮箱等，理想处理常常不是机械删除，而是在合适场景替换成连贯 synthetic information；但 Wikipedia 中公开人物姓名又可能是知识内容，不能一刀切。Representation 问题更根本：训练数据决定模型反映哪些声音、文化和观点；underrepresentation 会导致偏见、刻板印象和排斥，overrepresentation 会强化强势群体。African American dialect 的例子说明，如果模型只承认标准英语，它会把合法方言误判为错误。Copyright 页提醒：英国版权自动产生，保护原创且固定形式的作品，通常持续作者去世后 70 年；大多数互联网内容即使没标注也受保护。使用 copyrighted material 通常需要 license 或适用 UK fair dealing 的特定例外；license 还不等于平台 terms of use 允许，例如 Creative Commons 视频也可能受 YouTube 下载条款限制。

考试问法：如果考数据伦理，按 representation、toxicity、PII、copyright、license 五点答。高分答案要写出：数据选择不只是性能优化，它会改变模型公平性、安全性、隐私风险和合法性；能爬到的数据不等于可自由使用的数据。

考试速记版

- Data is the key: 模型只会从训练数据里学习知识、语言模式、推理模式和偏差。
- Base model = pre-training + annealing/mid-training; instruct model = post-training 后更会听指令的模型。
- Wikipedia 高质量、结构化、多语言、Creative Commons，但有地理和人口统计偏差。
- Web/Common Crawl 规模巨大、覆盖广，但噪声、重复、偏差和版权风险都很高。
- GitHub 数据对代码和结构化推理重要，license 与 permissively licensed code 是筛选重点。

- Data selection 目标: performance、data efficiency、selection efficiency、evaluation integrity、reduce bias/toxicity。
- Pretraining pipeline: language filtering → heuristics → quality/domain filtering → deduplication → toxicity/PII filtering → data mixing。
- Heuristic filtering 快但浅; classifier-based quality filtering 灵活但更贵。
- Deduplication 降成本、防过拟合、减少记忆重复内容; MinHash 用 signatures 近似 Jaccard similarity。
- FineWeb 的核心是 quality trumps quantity: 从 96 个 Common Crawl snapshots 构建约 15T token 数据并系统调 filters/dedup。
- FineWeb-Edu 用 Llama-3-70B-Instruct 给 460K 样本打教育价值 synthetic labels, 再训练轻量分类器过滤 15T tokens。
- Instruction tuning 用 instruction-response pairs 教模型理解任务和期望行为, 而不是只做 next-token continuation。
- Synthetic instruction data 有 self-generation + verification、distillation、evolutionary methods、procedural generation。
- Synthetic data 的风险: 质量控制、ground truth、teacher bias、多样性下降和 synthetic data collapse。
- 数据伦理要覆盖 representation、toxicity、PII、copyright、license 和 terms of use。

一段稳妥考场回答

Data is central to LLM development because a model can only learn the knowledge, language patterns, reasoning behaviours and biases present in its training data. Pre-training data builds a base model from large-scale raw text and code, mid-training or annealing can emphasise higher-quality or capability-specific data, and post-training data such as instruction-response pairs, RL and alignment data turns the model into an instruct model that follows human tasks and desired behaviours. Since raw sources such as Wikipedia, Common Crawl web data and GitHub differ greatly in quality, coverage, bias and legal constraints, modern LLM training depends on a data selection pipeline: language filtering, heuristic filtering, quality and domain filtering, deduplication, toxic/explicit and PII filtering, and data mixing. Deduplication, often using approximate methods such as MinHash, reduces repeated documents, training cost and memorisation. FineWeb illustrates that quality can beat quantity by systematically tuning filters and deduplication over Common Crawl; FineWeb-Edu further uses synthetic educational labels from a strong LLM to train a cheaper classifier for large-scale filtering. Finally, data selection is also an ethical and legal process: it affects representation, bias, toxicity, privacy, copyright and licensing, so good data engineering must optimise performance without pretending that filtering choices are neutral.